IEOR E4004: Project 2

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https://github.com/angelhuang1108/Scheduling-the-NBA

Question 1

For each team *i* we found the following information from the preliminary schedule:

- (a) All dates when team *i* played at home
- (b) For each opponent *j*, the number of times team *i* hosted *j* at home
- (c) For each opponent *j*, the number of times team *i* played away at *j*'s arena
- (d) All dates when team *i* played away

(a) & (d) Home and Away Dates

The table below lists each team's home game dates (a) and away game dates (d). The schedule spans November 1 – December 25, 2025 across 16 dates, with 8 games per date (128 total).

Team	(a) Home Dates	(d) Away Dates
Atlanta Hawks	Nov 03, Nov 07, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Nov 29, Dec 25	Nov 01, Nov 05, Nov 11, Nov 13, Nov 21, Dec 01
Boston Celtics	Nov 01, Nov 07, Nov 13, Nov 19, Nov 23, Nov 28	Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 21, Nov 27, Nov 29, Dec 01, Dec 25
Brooklyn Nets	Nov 01, Nov 05, Nov 11, Nov 13, Nov 21, Nov 29, Dec 01	Nov 03, Nov 07, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Dec 25
Chicago Bulls	Nov 01, Nov 03, Nov 13, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01	Nov 05, Nov 07, Nov 11, Nov 15, Nov 17, Nov 19, Dec 25
Cleveland Cavaliers	Nov 03, Nov 07, Nov 17, Nov 19, Nov 21, Nov 23, Nov 28, Nov 29, Dec 01, Dec 25	Nov 01, Nov 05, Nov 11, Nov 13, Nov 15, Nov 27
Dallas Mavericks	Nov 03, Nov 07, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Dec 25	Nov 01, Nov 05, Nov 11, Nov 13, Nov 21, Nov 29, Dec 01
Denver Nuggets	Nov 05, Nov 07, Nov 11, Nov 15, Nov 17, Nov 19, Nov 21, Dec 25	Nov 01, Nov 03, Nov 13, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01
Golden State Warriors	Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 27, Dec 25	Nov 01, Nov 07, Nov 13, Nov 19, Nov 21, Nov 23, Nov 28, Nov 29, Dec 01
Houston Rockets	Nov 03, Nov 11, Nov 13, Nov 15, Nov 19, Nov 21, Nov 23, Nov 27, Dec 25	Nov 01, Nov 05, Nov 07, Nov 17, Nov 28, Nov 29, Dec 01
Los Angeles Lakers	Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 19, Dec 25	Nov 01, Nov 07, Nov 13, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01
Miami Heat	Nov 01, Nov 05, Nov 13, Nov 17, Nov 27, Nov 28, Nov 29, Dec 01	Nov 03, Nov 07, Nov 11, Nov 15, Nov 19, Nov 21, Nov 23, Dec 25
Milwaukee Bucks	Nov 01, Nov 07, Nov 13, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01	Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 19, Dec 25
New York Knicks	Nov 01, Nov 05, Nov 11, Nov 13, Nov 15, Dec 01	Nov 03, Nov 07, Nov 17, Nov 19, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 25
Philadelphia 76ers	Nov 01, Nov 05, Nov 07, Nov 11, Nov 21, Nov 29, Dec 01	Nov 03, Nov 13, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Dec 25
Phoenix Suns	Nov 03, Nov 11, Nov 13, Nov 15, Nov 19, Nov 21, Nov 23, Dec 25	Nov 01, Nov 05, Nov 07, Nov 17, Nov 27, Nov 28, Nov 29, Dec 01
Toronto Raptors	Nov 01, Nov 05, Nov 07, Nov 17, Nov 27, Nov 28, Nov 29, Dec 01	Nov 03, Nov 11, Nov 13, Nov 15, Nov 19, Nov 21, Nov 23, Dec 25

(b) Home Matchup Counts: $c^H(i, j)$

Each cell shows the number of times the row team (*i*) hosted the column team (*j*) at home.

Team \ Opp.	ATL	BOS	BKN	CHI	CLE	DAL	DEN	GSW	HOU	LAL	MIA	MIL	NYK	PHI	PHX	TOR
Atlanta Hawks	—	1	0	1	0	1	0	1	0	1	1	1	1	0	1	1
Boston Celtics	0	—	0	0	0	1	0	1	0	0	1	1	0	1	1	0
Brooklyn Nets	1	1	—	0	0	1	1	1	0	0	1	0	0	1	0	0
Chicago Bulls	1	1	1	—	0	0	1	1	1	1	0	1	0	0	0	1
Cleveland Cavaliers	1	1	1	1	—	0	1	0	1	0	1	1	1	0	0	1
Dallas Mavericks	0	0	0	1	1	—	1	1	1	1	1	0	0	1	0	1
Denver Nuggets	1	1	1	0	0	0	—	0	1	1	1	0	1	1	0	0
Golden State Warriors	0	1	0	0	1	0	1	—	1	0	0	1	1	1	0	0
Houston Rockets	1	1	1	0	0	0	0	0	—	1	0	1	1	1	1	1
Los Angeles Lakers	0	1	1	0	1	0	0	1	0	—	0	0	1	0	1	1
Miami Heat	0	0	0	1	0	1	0	1	1	1	—	0	1	1	1	0
Milwaukee Bucks	0	0	1	0	1	1	1	0	0	1	1	—	1	1	1	0
New York Knicks	0	1	1	1	0	1	0	0	0	1	0	0	—	0	0	1
Philadelphia 76ers	1	0	0	1	1	0	0	0	0	1	0	0	1	—	1	1
Phoenix Suns	0	0	1	1	1	1	1	1	0	0	0	0	1	1	—	0
Toronto Raptors	0	1	1	0	0	0	1	1	1	0	1	1	0	0	1	—

(c) Away Matchup Counts: $c^A(i, j)$

Each cell shows the number of times the row team (*i*) played away at the column team's (*j*) arena.

Team \ Opp.	ATL	BOS	BKN	CHI	CLE	DAL	DEN	GSW	HOU	LAL	MIA	MIL	NYK	PHI	PHX	TOR
Atlanta Hawks	—	0	1	1	1	0	1	0	1	0	0	0	0	1	0	0
Boston Celtics	1	—	1	1	1	0	1	1	1	1	0	0	1	0	0	1
Brooklyn Nets	0	0	—	1	1	0	1	0	1	1	0	1	1	0	1	1
Chicago Bulls	1	0	0	—	1	1	0	0	0	0	1	0	1	1	1	0
Cleveland Cavaliers	0	0	0	0	—	1	0	1	0	1	0	1	0	1	1	0
Dallas Mavericks	1	1	1	0	0	—	0	0	0	0	1	1	1	0	1	0

Team \ Opp.	ATL	BOS	BKN	CHI	CLE	DAL	DEN	GSW	HOU	LAL	MIA	MIL	NYK	PHI	PHX	TOR
Denver Nuggets	0	0	1	1	1	1	—	1	0	0	0	1	0	0	1	1
Golden State Warriors	1	1	1	1	0	1	0	—	0	1	1	0	0	0	1	1
Houston Rockets	0	0	0	1	1	1	1	1	—	0	1	0	0	0	0	1
Los Angeles Lakers	1	0	0	1	0	1	1	0	1	—	1	1	1	1	0	0
Miami Heat	1	1	1	0	1	1	1	0	0	0	—	1	0	0	0	1
Milwaukee Bucks	1	1	0	1	1	0	0	1	1	0	0	—	0	0	0	1
New York Knicks	1	0	0	0	1	0	1	1	1	1	1	1	—	1	1	0
Philadelphia 76ers	0	1	1	0	0	1	1	1	1	0	1	1	0	—	1	0
Phoenix Suns	1	1	0	0	0	0	0	0	1	1	1	1	0	1	—	1
Toronto Raptors	1	0	0	1	1	1	0	0	1	1	0	0	1	1	0	—

Question 2

We wrote an Integer Program whose feasible solutions are all the valid schedules that preserve all structural properties of the original schedule while potentially reassigning which teams play each other on each date.

Sets and Indices

T = set of all teams ($|T| = 16$)

D = set of all game dates ($|D| = 16$)

$i, j \in T$ denote teams ($i \neq j$)

$d \in D$ denotes a game date

Parameters

From Question 1:

$H(i)$ = set of dates on which team i plays at home

$A(i)$ = set of dates on which team i plays away

$c^H(i,j)$ = number of times team i hosts team j

$c^A(i,j)$ = number of times team i visits team j's arena

Decision Variables

Define binary variables:

$$x_{ijd} \in \{0, 1\} \quad \text{for all} \quad i \neq j \in T,\ d \in D$$

where $x_{ijd} = 1$ if and only if team i hosts team j on date d.

Constraints

(e) Home Dates Preserved

For each team *i* and date *d*, team *i* plays exactly one home game if *d* is a home date, and none otherwise:

$$\Sigma_{j \neq i}\ x_{ijd} = 1\ [d \in H(i)] \quad \text{for all } i \in T,\ d \in D$$

On each of team *i*'s original home dates, exactly one opponent visits; on all other dates, no home game is played.

(f) Away Dates Preserved

For each team *i* and date *d*, team *i* plays exactly one away game if *d* is an away date, and none otherwise:

$$\Sigma_{j \neq i}\ x_{jid} = 1\ [d \in A(i)] \quad \text{for all } i \in T,\ d \in D$$

On each of team *i*'s original away dates, team *i* visits exactly one opponent; on all other dates, no away game is played.

(g) Home Matchup Counts Preserved

Over all dates, team *i* hosts team *j* exactly $c^H(i,j)$ times:

$$\Sigma_{d \in D}\ x_{ijd} = c^H(i,j) \quad \text{for all } i \neq j \in T$$

The total number of home games team *i* plays against each specific opponent *j* is preserved from the original schedule.

(h) Away Matchup Counts Preserved

Over all dates, team *i* visits team j's arena exactly $c^A(i,j)$ times:

$$\Sigma_{d \in D}\ x_{jid} = c^A(i,j) \quad \text{for all } i \neq j \in T$$

The total number of away games team *i* plays at team *j*'s arena is preserved from the original schedule.

Complete Formulation

The full IP (without objective) is:

Find $x_{ijd} \in \{0,1\}$ for all $i \neq j \in T,\ d \in D$

subject to:

$\Sigma_{j \neq i}\ x_{ijd} = 1[d \in H(i)]$ for all $i \in T$, $d \in D$ (e)

$\Sigma_{j \neq i}\ x_{jid} = 1[d \in A(i)]$ for all $i \in T$, $d \in D$ (f)

$\Sigma_{d \in D}\ x_{ijd} = c^H(i,j)$ for all $i \neq j \in T$ (g)

$\Sigma_{d \in D}\ x_{jid} = c^A(i,j)$ for all $i \neq j \in T$ (h)

$x_{ijd} \in \{0,1\}$ for all $i \neq j \in T$, $d \in D$

Model Size and Result

Variables: 16 × 15 × 16 = 3,840 binary variables

Constraints: 2 × 16 × 16 + 2 × 16 × 15 = 992 linear constraints

Running our model with Gurobi returned "Status: Optimal", confirming the constraint system is consistent and at least one valid reassignment of opponents exists.

Question 3

Additional Constraint

Building on the feasibility model of Question 2, we add the constraint that limits how many time zones a team must cross over any three consecutive games. For each team *t* and each triple of consecutive game dates (d_1, d_2, d_3) in its schedule, the sum of consecutive time zone differences must not reach 4 or more:

$$|tz(t,d_2) - tz(t,d_1)| + |tz(t,d_3) - tz(t,d_2)| \leq 3 \quad \text{for all } t \in T, \text{ consecutive triples } (d_1,d_2,d_3)$$

where tz(t, d) denotes the UTC offset of the arena where team *t* plays on date *d*. Each arena is assigned to one of four time zone groups: Eastern (UTC−5), Central (UTC−6), Mountain (UTC−7), or Pacific (UTC−8).

Linearization

Since tz(t, d) is itself a linear expression in the x variables (each team plays in exactly one arena per game date), the absolute-value terms are linearized in the standard way. Let $\delta_{12} = tz(t,d_2) - tz(t,d_1)$ and $\delta_{23} = tz(t,d_3) - tz(t,d_2)$. We introduce auxiliary variables $p_{12}, p_{23} \geq 0$ for each (team, triple) pair:

$$p_{12} \geq \delta_{12}, \quad p_{12} \geq -\delta_{12} \quad \Rightarrow \quad p_{12} = |\delta_{12}|$$

$$p_{23} \geq \delta_{23}, \quad p_{23} \geq -\delta_{23} \quad \Rightarrow \quad p_{23} = |\delta_{23}|$$

$$p_{12} + p_{23} \leq 3$$

Result (Infeasible)

The model is infeasible. Gurobi proves that no schedule satisfying all constraints (e)–(h) from Question 2 can simultaneously satisfy the time zone constraint for every team.

To diagnose this, we computed the IIS (Irreducible Inconsistent Subsystem), the smallest subset of constraints that is still collectively infeasible. The IIS Gurobi returns contains 31 structural constraints from (e)–(h) together with 3 time zone constraints, and all 3 time zone constraints involve a single team, the Golden State Warriors, on triples k = 2, 4, and 10 of its game sequence. This makes sense since the time zone difference from Pacific (UTC−8) to Eastern (UTC−5) is 3 hours, so any back-to-back crossing of those boundaries produces a sum of 4 or more, and Golden State's home/away pattern forces several such crossings.

Relaxation with Slack Variables

Since the hard constraint is infeasible, we relax it by introducing a non-negative slack variable *s(t, k)* for each team *t* and consecutive triple *k*. The constraint becomes:

$$p_{12} + p_{23} \leq 3 + s(t,k), \quad s(t,k) \geq 0$$

We then minimize the total slack across all teams and triples:

$$\min \ \Sigma_{t,k} \ s(t,k)$$

This finds the schedule that comes closest to satisfying the time zone constraint (the one that violates it by the least possible total amount). The optimal total slack is **27** hours, distributed across 9 teams and 19 triples. Golden State Warriors accounts for the largest single share (10 hours across 5 triples), followed by Phoenix Suns (4 hours, 3 triples) and the Lakers (3 hours, 3 triples). The remaining 6 teams contribute 1–2 hours each. By comparison, the original draft schedule has 22 violating triples and 33 hours of total excess, so the IP reduces unavoidable time zone burden by 6 hours (≈ 18%) without changing any team's home/away dates or matchup counts. The 27-hour figure is also a lower bound:

no schedule preserving (e)–(h) can achieve a smaller total violation. The corresponding schedule can be found in the file *schedule_q3_relaxed.csv*.

Question 4

For Question 4, we considered two different ways to improve the current schedule. Method 1 minimizes the total distance travelled across all teams. Method 2 minimizes the maximum deviation from the league-average travel distance.

Computing Travel Distance

For each method, travel distances are computed as Haversine distances in miles between consecutive arena locations for each team. For team *t* travelling between consecutive game dates d_1 and d_2, four cases arise depending on whether each game is home or away:

Home → Home (HH): zero travel (both games are at the team's own arena).

Home → Away (HA): dist(arena(t), arena(h_2)) where h_2 is the host on d_2. This is linear in x.

Away → Home (AH): dist(arena(h_1), arena(t)) where h_1 is the host on d_1. Also linear in x.

Away → Away (AA): dist(arena(h_1), arena(h_2)), which is a product of two binary variables (bilinear).

Linearization of Away → Away Legs

The bilinear AA terms are linearized by introducing auxiliary binary variables w(t, d_1, d_2, h_1, h_2) ∈ {0, 1} for each (team, date pair, host pair) combination:

$$w \leq x_{h_1,t,d_1}, \quad w \leq x_{h_2,t,d_2}, \quad w \geq x_{h_1,t,d_1} + x_{h_2,t,d_2} - 1$$

These three constraints enforce w = x(h_1,t,d_1) · x(h_2,t,d_2), the standard linearization of a binary product. The travel contribution of each AA leg is then dist(arena(h_1), arena(h_2)) · w, which is linear.

Method 1 – Minimizing Total Travel Distance

$$\min \ \Sigma_{t \in T} \ \text{travel}(t) \quad \text{subject to constraints (e)–(h)}$$

This objective targets league-wide efficiency: the total miles flown across all 16 teams. We believe that this could be a solid improvement from the league's perspective, or from an environmentally friendly point of view, as it would likely minimize the total travel cost and time.

Model Size

Variables: 3,840 x-vars + ~13,275 w-vars = ~17,115 binary variables

Constraints: 992 base + 3 × (number of AA legs) linearization constraints for w

Results

The total travel leaguewide went from 200,087 miles in the original schedule to 192,082 miles, a 4% reduction. The maximum team travel under this objective is 20,517 miles (Golden State, essentially unchanged from the original 20,400). This is expected: with the total-travel objective, the model has no incentive to reduce any single team's travel beyond what helps the sum, so individual teams may go up or down. Toronto and Brooklyn each gain ~1,300 miles while Boston, Cleveland, and several others lose more than 1,000. This optimized schedule can be found in *schedule_min_total.csv*.

Travel comparison (original vs. optimized):

Team	Original (mi)	Optimized (mi)	Change (mi)
Golden State Warriors	20,400	20,517	+117
Phoenix Suns	16,730	15,971	-759
Los Angeles Lakers	16,166	15,199	-967
Boston Celtics	15,756	13,520	-2,236
New York Knicks	15,669	14,637	-1,032
Philadelphia 76ers	13,977	12,864	-1,113
Dallas Mavericks	13,573	13,311	-262
Miami Heat	12,798	12,503	-295
Brooklyn Nets	12,385	13,630	+1,245
Toronto Raptors	10,789	12,134	+1,345
Denver Nuggets	9,922	8,756	-1,166
Cleveland Cavaliers	9,688	8,320	-1,368
Houston Rockets	9,204	8,528	-676
Milwaukee Bucks	8,458	8,258	-200
Atlanta Hawks	7,308	7,234	-74
Chicago Bulls	7,265	6,701	-564
TOTAL	**200,087**	**192,082**	**-8,005**

Method 2 – Minimizing Maximum Deviation from League-Average Travel

As an alternative improvement, we also solved a model that minimizes the maximum deviation of any team's total travel distance from the league-average travel distance. Let $travel(t)$ denote the total travel distance of team t, and let $\bar{D}$ denote the average travel distance across all teams. Introducing a scalar variable δ, the objective is:

$$\mathrm{Min}(\delta)$$

subject to:

$$\delta \geq travel(t) - \bar{D} \quad \forall t \in T$$
$$\delta \geq \bar{D} - travel(t) \quad \forall t \in T$$

where:

$$\bar{D} = \frac{1}{|T|} \sum_{t \in T} travel(t)$$

This is a min-max fairness formulation: δ measures the largest absolute deviation from the league-average travel, and minimizing it makes team travel as balanced as possible.

Model Size

Variables: 3,840 x-vars + ~13,275 w-vars + 16 travel-vars + 1 δ= ~17,132 binary/continuous variables

Constraints: 992 base + 16 travel-definition constraints + 32 fairness-deviation constraints + w linearization constraints

Results

Under this fairness-based objective, the optimized schedule achieves a league-average travel distance of 13,140 miles and reduces the maximum deviation from the league average from 7,895 miles to 5,000 miles, which corresponds to a **36.7%** improvement in fairness. The travel range is also reduced from 13,135 miles to 9,882 miles. This optimized schedule can be found in *schedule_min_dev.csv*.

Team-by-Team Travel Comparison: Original vs. Optimized Schedule

Team	Original	Optimized	vs Orig Avg	vs Opt Avg
Golden State Warriors	20,400	18,045	+7,895	+4,905
Phoenix Suns	16,730	15,532	+4,225	+2,392
Los Angeles Lakers	16,166	14,909	+3,660	+1,769
Boston Celtics	15,756	18,140	+3,251	+5,000
New York Knicks	15,669	17,681	+3,163	+4,542
Philadelphia 76ers	13,977	14,614	+1,472	+1,474
Dallas Mavericks	13,573	13,390	+1,067	+251
Miami Heat	12,798	12,986	+292	-154
Brooklyn Nets	12,385	13,881	-120	+741
Toronto Raptors	10,789	11,665	-1,716	-1,474
Denver Nuggets	9,922	9,376	-2,583	-3,764
Cleveland Cavaliers	9,688	13,727	-2,817	+587
Houston Rockets	9,204	11,274	-3,302	-1,866
Milwaukee Bucks	8,458	8,258	-4,048	-4,881
Atlanta Hawks	7,308	8,313	-5,198	-4,827
Chicago Bulls	7,265	8,443	-5,241	-4,697
League average	12,505	13,140		

Fairness Summary for the Original and Fairness-Optimized Schedules

Metric	Original	Optimized	Improvement
Max deviation from avg (δ)	7,895	5,000	+2,894
Range (max - min travel)	13,135	9,882	+3,254
Max travel	20,400	18,140	+2,260
Min travel	7,265	8,258	-994
Total travel	200,087	210,232	-10,146

Method 1 and Method 2 improve the schedule according to different objectives. Method 1 minimizes total leaguewide travel, prioritizing efficiency: it cuts 8,005 miles (≈ 4%) from the original 200,087-mile total, but with the Warriors' worst-case essentially unchanged (20,400 → 20,517) and several teams traveling more than they did before. Method 2 minimizes the maximum deviation from the league average, prioritizing fairness: it compresses the spread (max deviation 7,895 → 5,000 mi, range 13,135 → 9,882 mi) and brings Golden State's travel down by over 2,000 miles, but at a 5% increase in total leaguewide travel (200,087 → 210,232 mi). The two methods produce different schedules and help different teams, which highlights an efficiency–fairness tradeoff. Minimizing total miles traveled doesn't balance the load, and balancing the load will cost the league miles. Depending on the league's main objective however, we believe one of these improvements could definitely help optimize the schedule.